

Xinyue (Margaret) Zhang

☎ (657) 793-5042 ✉ xinyuezhang2025@u.northwestern.edu 🔗 linkedin.com/in/xinyue-zhang-zxy 📁 github.com/xinyue0124

Education

Northwestern University

Evanston, IL

M.S. in Computer Science — GPA: 3.67/4.0

Sep 2025 – Expected Jun 2027

– Coursework: Deep Learning for NLP, Machine Learning, Scalable Software Architecture, Formal Verification

University of Wisconsin – Madison

Madison, WI

B.S. in Computer Science — GPA: 3.66/4.0 · Dean's List 2021–2024

Sep 2021 – Dec 2024

– Coursework: Database Management Systems, Big Data Systems, Data Pipeline, Algorithms, Probability Theory

Professional Experience

Guoke Hengtai (Beijing) Medical Technology Co., Ltd.

Beijing, China

Software Developer Intern

Jan 2025 – Aug 2025

- Optimized warehouse inventory queries on a medical-device platform by redesigning MyBatis/MySQL composite indexes, supporting **5,000+ daily inventory requests** while reducing unnecessary database work.
- Implemented a **Redis cache-aside** data-access layer in Spring Boot using AOP **@Cacheable** to serve frequently accessed inventory records from cache and reduce repeated reads against the transactional database during peak workloads.
- Diagnosed lock-timeout failures on hot inventory records under concurrent updates; redesigned transaction locking around **two-phase locking** and validated behavior with JUnit tests at **200+ concurrent threads** to improve stability under write contention.

Chanmao Song Network Technology Co., Ltd.

Urumqi, China

Software Developer Intern

May 2024 – Aug 2024

- **Led a 5-person team** embedded with the client, combining stakeholder interviews with session-log analysis to identify **three high-friction workflows** and translate findings into a prioritized delivery roadmap.
- Delivered redesigned workflows across Vue.js and REST API layers and validated releases through **A/B testing**, reducing interaction steps by **30%** and increasing 7-day retention by **40%**.
- Built a modular API layer and JavaScript test suite covering **12 critical user paths** across Agile releases, improving release reliability while contributing to **15% post-launch revenue growth**.

Projects

Automated TA | Node.js, React/TypeScript, PostgreSQL, pgvector, Docker

Sep 2025 – Present

- Owned an AI-powered application **end-to-end** in React/TypeScript and Node.js, operating in **production for 200+ students** with a backend designed for continuous 24/7 availability.
- Architected **multi-tenant REST APIs** on **PostgreSQL** with JWT authentication, rate limiting, and user-scoped queries, enforcing data isolation across users at the application and query layers.
- Built the **retrieval data pipeline** from document chunking through **pgvector** cosine-similarity search and top-3 retrieval, and containerized the service with **Docker** for reproducible deployment and CI/CD readiness.

NYC Taxi ELT Pipeline | Airbyte, Snowflake, dbt, SQL, Parquet

Sep 2024 – Dec 2024

- Built an **ELT pipeline** for NYC taxi trip data, using **Airbyte** to load trip records from Parquet and CSV files and articles from a news API into **Snowflake** as raw tables.
- Modeled the raw trips with **dbt** SQL transformations, classifying every pickup and dropoff location ID into one of seven NYC districts so trips could be analyzed by area instead of by raw ID.
- Queried the modeled tables in Snowflake with **SQL** to compare pickup and dropoff volume across districts, showing where trips start versus where they end.

Serving Under Load | gRPC, HDFS, Spark, Kafka, Python, Docker

Jan 2024 – May 2024

- Built a gRPC key-value service with protobuf schemas, a thread-pool server, and an LRU result cache, separating request handling from repeated computation to support concurrent workloads.
- Implemented multithreaded lock-based synchronization across systems workloads, strengthening concurrency control and resource efficiency under parallel execution.
- Built distributed data-processing workflows across HDFS, Spark RDDs/DataFrames/SQL, and Kafka producers/consumers, covering storage, batch computation, and streaming data movement.

Technical Skills

Languages: Python, Java, C/C++, JavaScript/TypeScript, C#, SQL

Databases: MySQL, PostgreSQL, pgvector, Redis, NoSQL, MongoDB, Cassandra, Elasticsearch, BigQuery, Snowflake

Data Pipelines: HDFS, MapReduce, Spark, Kafka, Airbyte, dbt, ETL/ELT

Backend: React, Node.js, Spring Boot, REST/GraphQL, gRPC, Docker

Cloud & AI: AWS (S3, RDS, SES, Rekognition, Beanstalk), GCP, PyTorch, LoRA/PEFT, RAG

Systems: Distributed Systems, Concurrency Control, Database Internals, TLA+, Linux, Git, CI/CD, JUnit